Second Research of GPT 5.6 High Sol

Mechanistic estimation of deep third-cumulant content in random ReLU MLPs

Scope. Second-round analysis after the measured closure of low-rank product sketches, self-averaging, source dropping, slice-only closures, Tucker/CP routes, Strassen, sampling hybrids, and learned corrections. Claims are labeled Derived, Conjecture, or Measure-this.

Q1 - Aggregate response state: the surviving idea

[Derived] Representation. Write the deep third-cumulant tensor as a block-term / one-mode-separable approximation

$$K_{old} \approx \Sigma_{a=1}^{k} R(G_a, v_a), \quad R(G,v) = \mathrm{sym}(G \otimes v),$$

with $G_a \in R^{n\times n}$ symmetric and $v_a \in R^n$. Under linear transport $B = W \mathrm{diag}(w1)$, the family is closed exactly:

$$G_a' = B\, G_a\, B^T, \qquad v_a' = B\, v_a.$$

Its required slices are

$$D21_{ic} = \Sigma_a [(G_a)_{ii} (v_a)_c + 2 (G_a)_{ic} (v_a)_i],$$

$$D3_i = 3\, \Sigma_a (G_a)_{ii} (v_a)_i.$$

One component can already generate a full-rank D21 because G_a is dense. Therefore the near-full spatial rank of $A \circ P$ and the ≈ 230 spatial rank needed for 2% error in U do not imply $k \approx 230$. The relevant quantity is separation rank between the symmetric two-index block and the third index.

Concrete fitting procedure

For a source at the age gate,

$$K_s = \Sigma_j d_j\, \mathrm{sym}(a_j\, a_j^T \otimes p_j), \quad d_j = w2_j.$$

Use a two-stage fit.

Stage 1 - block-term SVD initializer. Matricize the unsymmetrized tensor as (12)|3:

$$M = \Sigma_j d_j\, \mathrm{vec_s}(a_j\, a_j^T)\, p_j^T = X\, P^T.$$

The right Gram can be formed without materializing the $n^2 \times n$ matrix:

$$M^T M = P\, K_X\, P^T, \qquad (K_X)_{jl} = d_j\, d_l\, (a_j^T a_l)^2.$$

Take the top k eigenvectors v_a of this $n\times n$ Gram. Recover each dense G_a from

$$G_a \propto A\, \mathrm{diag}(d \odot P^T v_a)\, A^T.$$

[Derived cost] A^T A costs ≈1 unit. The two dense multiplies for P K_X P^T cost ≈2 units. Constructing k dense G_a matrices costs ≈k units. Initial fitting is therefore about (3+k) units once at the gate. This is too expensive if repeated blindly in production, but ideal for the first falsification experiment.

Stage 2 - query-weighted refinement. Refine the factors directly against the four existing query channels:

$$\min_{G_a,v_a} \Sigma_{m=1}^4 \|U_m - Q_m(\Sigma_a R(G_a,v_a))\|_F^2 / \|U_m\|_F^2.$$

For fixed v_a, the problem is linear in G_a; use variable projection or CG, then update v_a with a small nonlinear optimizer. Because evaluating the slices of R(G,v) is O(k n^2), direct slice fitting is cheap. If the n×384 intermediates U_m must be reproduced explicitly, fit on a fixed 64-128 column sketch and validate on the full U_m.

[Cost] A slice-only refinement iteration is roughly 10-50 k n^2 FLOPs, i.e. about 0.005-0.025 k units/iteration. A 128-column projected-U refinement remains below about 0.2 k units/iteration.

What k is economically plausible?

[Conjecture] Current-layer query error may need only k=1-4 even though each U has high spatial rank. Future-stable representation is probably k=2-6. But transport dominates:

$$\text{transport} \approx 2k \text{ units/layer.}$$

- k=1: ≈20 units over ten deep layers.
- k=2: ≈40 units.
- k=4: ≈80 units.
- k=6: ≈120 units.

Thus k>4 is not economically interesting, and a leaderboard-compatible mechanism strongly points to k≤2, or to an algebraically equivalent state with the same effective response rank.

First falsifier

[Measure-this] On one stored age-4 source (.npz legs available), fit k=1,2,4,8. Measure all four full query errors at the gate. Then transport teacher and fitted tensor through the actual next one and three B matrices, without refitting, and re-measure the four queries.

Kill the idea if k>4 is needed for <2% at the gate, or if k≤2 gives <2% initially but exceeds roughly 4% after one transport. This is the highest-value experiment in this round and is comfortably under one hour on a CPU for one source/layer.

Q2 - Other genuinely source-free formulations

1. Finite-horizon observability / balanced truncation

[Conjecture - credible] Treat deep transport as a linear time-varying system

$$K_{l+1} = B_l^{\otimes 3} K_l + S_l, \qquad y_l = Q_l K_l,$$

where y_l denotes the four query outputs. Instead of preserving K_l, preserve only coordinates in the subspace observable along the actual remaining quenched weight sequence:

$$z_{l+1} = A_l^{(r)} z_l + b_l, \qquad y_l \approx \Sigma_{a=1}^{k} z_{la} Y_{la}.$$

This is not the already-dead adjoint idea: the old sources are merged into one reduced global state rather than contracted one source at a time. The relevant rank is a finite-horizon Hankel/observability rank, not the instantaneous spatial rank of U.

[Cost] If Y_{la} are dense n×n output templates, online reconstruction is $O(k\, n^2)$, below 0.001k units/layer. A naive backward construction of block-dual modes is roughly 2k units/layer, with birth projection at most another k units/layer. It is only interesting for $k \le 2$.

[First falsifier] At layers 10 and 14, stack the four channel-normalized query outputs from all old sources as samples and compute an SVD across sources. If the first two cross-source modes do not explain at least 99.96% of squared energy (the 2% RMS target), a two-state observability model is dead immediately. This should run in minutes from stored query matrices.

2. Conditional odd-Wiener-chaos model

[Conjecture - lower probability, but not ruled out by parity alone] T1 proves $E[y \mid \text{even statistics}] = 0$. It does not prove that y is unpredictable from the realized odd current-layer weight W_l. For fixed incoming K,

$$(K')_{iic} = \Sigma_{abd} B_{ia} B_{ib} B_{cd} K_{abd}.$$

Decompose $B_i B_i^T$ into its covariance plus a centered quadratic fluctuation. The covariance term gives a first-Hermite contribution

$$(K')^{(1)}_{iic} = \Sigma_d B_{cd} t_d, \qquad t_d = \Sigma_{ab} (\Sigma_B)_{ab} K_{abd}.$$

If T2 is exactly the projection onto this first-Wiener-chaos term, then its 12-16% floor already kills the zeroth-order version. The only genuinely new hypothesis is that, after aggregation over all old sources, the residual Hermite-3 component has a very small conditional covariance rank.

Carry a partial-trace vector t_l plus m coefficients of residual quenched modes. The slice realization is $O(m\, n^2)$, effectively <0.01 units/layer for $m \le 8$, provided the residual templates themselves admit a cheap recurrence.

[First falsifier] Freeze one pre-layer old state. On 128 selected output rows, resample the current Gaussian W_l 32-64 times, subtract the analytic first-Hermite component, and PCA the residual query responses. Kill the idea unless the first $m \le 8$ modes retain >99.96% of residual energy. My prior is that this fails.

3. Second-order chaos / Hessian-state formulation

[Derived rejection] Approximate each hidden fluctuation by

$$\delta h_i \approx J_i \xi + \xi^T H_i \xi - \mathrm{tr}(H_i).$$

Then

$$J' = D1\, W\, J,$$

$$H'_i = D1_i \Sigma_j W_{ij} H_j + D2_i (WJ)_i (WJ)_i^T.$$

This is genuinely source-free: new births are automatically summed into H. But with chaos dimension m, storing H costs $n\, m^2$ and transporting its output index costs about $2\, n^2\, m^2$ FLOPs, i.e.

$$\text{cost} \approx m^2 / n \text{ units/layer.}$$

- m=100: ≈9.8 units/layer.
- m=150: ≈22 units/layer.

M1/M3 make $m \ll 100$ unlikely. This cannot explain the leaders unless a new deterministic chaos basis gives $m \lesssim 40$.

[First falsifier] Project the stored old-source quadratic response onto the best m=32,64,96 deterministic chaos directions. If m=64 is not already below 2% query error, stop. No full chain run is needed.

Q3 - Most likely explanation of the leaders

[Conjecture] The most likely mechanism is a very-low separation-rank aggregate response state, probably k=1 or 2, or an algebraically equivalent finite-horizon response compression.

Why this matches the evidence:

- Deep-tier cost becomes independent of the number of old sources.
- Dense G can generate a high-spatial-rank D21, so M1/M3 do not rule out small separation rank.
- Linear transport of R(G,v) is exact.
- k=1-2 costs only about 20-40 units over roughly ten deep layers, consistent with a 133-165-unit total if their young tier is also somewhat better optimized.
- It naturally explains why the leaders pay approximately the young-tier bill rather than a cost growing with old-source count.

[Measure-this] The distinguishing observable is cross-source / temporal response rank, not singular rank of U. At layers 10 and 14, stack the four normalized query outputs from every old source and compute the SVD across sources. If the first 1-2 response modes explain >99.96% of squared energy, the leaderboard mystery has a plausible mechanism. If more than four modes are required, the aggregate-response route is economically weak; given T1-T3 and M1-M4, there is then no currently credible O(n^2)-per-layer representation left.

Single highest-value next experiment

Fit k=1,2,4,8 aggregate response components to one stored age-4 source, validate the four query channels at the gate, then propagate teacher and approximation through one and three actual future B matrices without refitting. This experiment simultaneously tests separability, future stability, and the only remaining cost regime compatible with the leaderboard.